

Lab 05 — Databases & Drizzle ORM

PB138

ERD • relationships • migrations • seeding • access-scoped queries

Goals

- ORM
- Migrations
- Drizzle
- Faker.js
- Modeling
- Assignment!

Quick recap

- **Primary Keys** → UUID/serial
- **Constraints** → NOT NULL, UNIQUE, FK
- **Normalization** → less duplication

ERD — core rules

- Entity = table
- Attribute = column
- **1:N** → FK on the *many* side
- **M:N** → always a join table
- Never store arrays of IDs in a column

Primary keys

UUID	serial
Hard to guess	Sequential
Suitable for public IDs	Mostly internal

Recommendation: UUID for web applications

Soft delete

```
UPDATE tasks SET deleted_at = NOW() WHERE id = ?;
```

- Data stays recoverable
- Foreign keys remain intact
- Always filter with `deleted_at IS NULL`

Object Relation Mapper (ORM)

- A software connecting DBs with object-oriented languages
- Simplifies DB operations by mapping tables into language native objects
- Reduces need for raw SQL queries
- Improves readability, maintainability and security

Migrations

- Version control for your database schema.
- Track changes to tables, columns, and relations.
- Saved as timestamped files and tracked in **Git**.
- Allow rolling back to previous schema versions

Drizzle ORM (TypeScript)

Key Features

- **Type-safe queries** using TypeScript
- Works with **PostgreSQL, MySQL, SQLite**
- Supports **migrations**
- Lightweight and fast
- Close to **raw SQL syntax**

```
import { db } from "../db";  
import { users } from "../schema";  
  
const allUsers = await db.select().from(users);
```

Drizzle Schema

- Full type safety
- Autocompletion in IDE
- Single source of truth for database structure

Example Schema

```
import { pgTable, serial, text, integer } from "drizzle-orm/pg-core";

export const users = pgTable("users", {
  id: serial("id").primaryKey(),
  name: text("name").notNull(),
  age: integer("age")
});
```

Drizzle Studio

- Allows connecting and interacting with the database based on the drizzle schema

The screenshot displays the Drizzle Studio application interface. On the left, a sidebar contains a 'SQL runner' and a 'Drizzle runner' section. Below these, a 'Search tables' input is followed by a list of database tables: Customers (93 records), Employees (9), Order Details (2155), Orders (830), Products (77), Shippers (3), and Suppliers (29). The 'Customers' table is selected and highlighted. The main area shows a table view of the 'Customers' data. At the top of this view, there are controls for 'Filters', 'Columns', and an 'Add record' button. To the right, it indicates '50 records' and provides navigation controls (previous, next, first, last, search, and download). The table itself has columns for CustomerID, CompanyName, ContactName, ContactTitle, Address, City, PostalCode, and Region. The first few rows of data are visible, showing various customer records from different regions.

CustomerID	CompanyName	ContactName	ContactTitle	Address	City	PostalCode	Region
ALFKI	Alfreds Futterkist...	Maria Anders	Sales Representative	Obere Str. 57	Berlin	12209	Western Euro
ANATR	Ana Trujillo Empar...	Ana Trujillo	Owner	Avda. de la ...	México D.F.	05021	Central Amer
ANTON	Antonio Moreno Taq...	Antonio Moreno	Owner	Mataderos 23...	México D.F.	05023	Central Amer
AROUT	Around the Horn	Thomas Hardy	Sales Representative	120 Hanover ...	London	WA1 1DP	British Isle
BERGS	Berglunds snabbköp	Christina Berglund	Order Administrator	Berguvsvägen...	Luleå	S-958 22	Northern Eur
BLAUS	Blauer See Delikat...	Hanna Moos	Sales Representative	Forsterstr. ...	Mannheim	68306	Western Euro
BLONP	Blondesddsl père e...	Frédérique Citeaux	Marketing Manager	24, place Kl...	Strasbourg	67000	Western Euro
BOLID	Bólido Comidas pre...	Martín Sommer	Owner	C/ Araquil, ...	Madrid	28023	Southern Eur
BONAP	Bon app'	Laurence Lebihan	Owner	12, rue des ...	Marseille	13008	Western Euro
BOTTM	Bottom-Dollar Mark...	Elizabeth Lincoln	Accounting Manager	23 Tsawassen...	Tsawassen	T2F 8M4	North Americ
BSBEV	B's Beverages	Victoria Ashworth	Sales Representative	Fauntleroy C...	London	EC2 5NT	British Isle
CACTU	Cactus Comidas par...	Patricio Simpson	Sales Agent	Cerrito 333	Buenos Aires	1010	South Americ
CENTC	Centro comercial M...	Francisco Chang	Marketing Manager	Sierras de G...	México D.F.	05022	Central Amer
CHOPS	Chop-suey Chinese	Yang Wang	Owner	Hauptstr. 29	Bern	3012	Western Euro
COMMI	Comércio Mineiro	Pedro Afonso	Sales Associate	Av. dos Lusi...	Sao Paulo	05432-043	South Americ
CONSH	Consolidated Holdi...	Elizabeth Brown	Sales Representative	Berkeley Gar...	London	WX1 6LT	British Isle
DRACD	Drachenblut Delika...	Sven Ottilieb	Order Administrator	Walserweg 21	Aachen	52066	Western Euro
DUMON	Du monde entier	Janine Labrune	Owner	67, rue des ...	Nantes	44000	Western Euro
EASTC	Eastern Connection	Ann Devon	Sales Agent	35 King Geor...	London	WX3 6FW	British Isle
ERNSH	Ernst Handel	Roland Mendel	Sales Manager	Kirchgasse 6	Graz	8010	Western Euro
FAMIA	Familia Arquibaldo	Aria Cruz	Marketing Assistant	Rua Orós, 92	Sao Paulo	05442-030	South Americ
FISSA	FISSA Fabrica Inte...	Diego Roel	Accounting Manager	C/ Moralzarz...	Madrid	28034	Southern Eur
FOLIG	Folies gourmandes	Martine Rancé	Assistant Sales Age...	184, chaussé...	Lille	59000	Western Euro

Faker.js

- A library to generate fake data for testing & development.
- Provides utilities for generating common data formats

```
import { faker } from "@faker-js/faker";

const user = {
  name: faker.person.fullName(),
  email: faker.internet.email(),
};
```

PlantUML

- free tool for creating UML diagrams through plain text language
- enables easy version control
- used for various diagrams
 - ERD
 - Class
 - State
 - ...

@startuml

```
entity Order {
  +id : <<uuid>>
  ---
  created_at : timestamp
  status : text
}
```

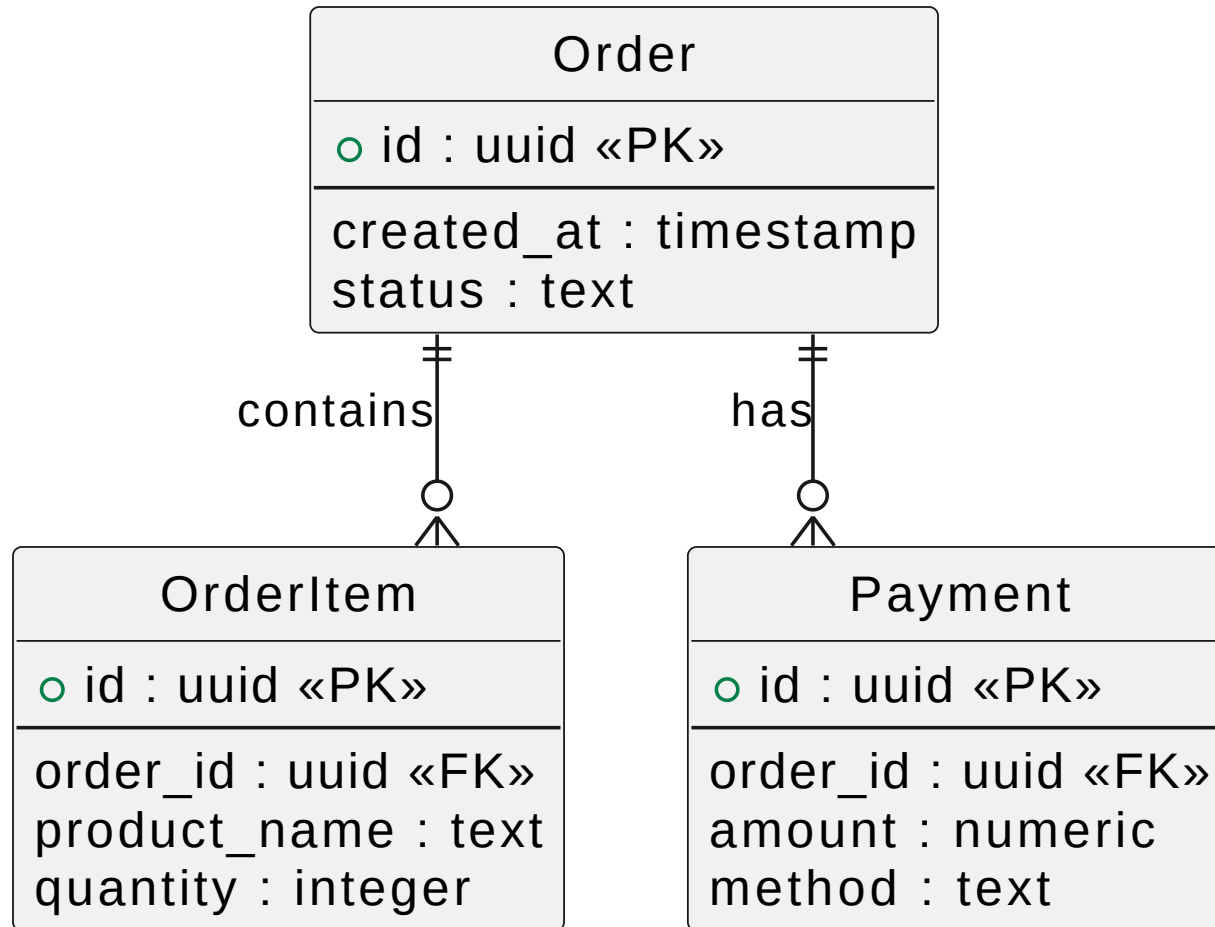
```
entity OrderItem {
  +id : <<uuid>>
  ---
  order_id : <<uuid FK>>
  product_name : text
  quantity : integer
}
```

```
entity Payment {
  +id : <<uuid>>
  ---
  order_id : <<uuid FK>>
  amount : numeric
  method : text
}
```

```
Order ||--o{ OrderItem : contains
Order ||--o{ Payment : has
```

@enduml

Solution



To the assignment!

You can find assignment in the interactive syllabus